ENME608: Mechanism and Dynamics of Machinery

Lecture 1: Introduction and Fundamentals

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Assessment

Assessment Type	Weighting	Due Date
Mid-semester Test	25%	In Week 6 (Date & Time: TBA)
Assignment	25%	31 May 2026, Online Submission by 11:59 PM
Final Controlled Assessment	50%	After Week 12 (Date & Time: TBA)

Reference

- Recommended Textbook
  - Robert L. Norton, *Design of Machinery : An Introduction to the Synthesis and Analysis of Mechanisms and Machines,* 6th Edition, McGraw Hill, 2019, ISBN-10: 1260113310 , ISBN-13: 978-1260113310

    (Online version available: AUT Library or

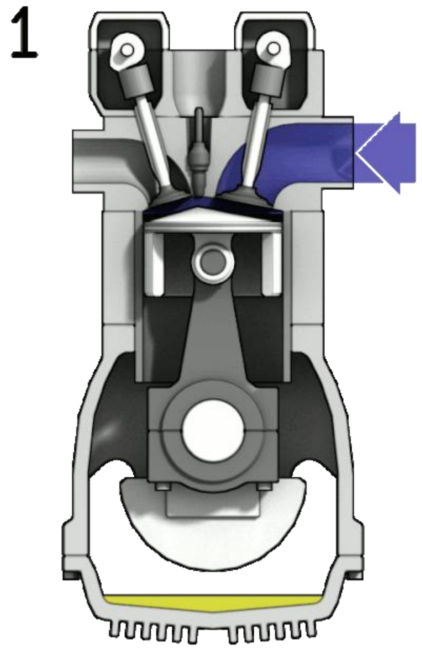
    ENME608: Course Resources in Canvas)

Other References

- THEORY OF MACHINE AND MECHANISMS Joseph Edward Shigley and John Joseph Uicker, JR. McGraw-Hill International Editions 1995 2.
- KINEMATICS AND DYNAMICS OF MACHINERY Charles E. Wilson and J. Peter Sadler 3rd Edition
- MECHANISM ANALYSIS Lyndon O. Barton Second Edition Marcel Dekker, Inc. 1993

What is ENME608?

- A branch of Applied Mechanics
- Relationship between the geometry and motions of the parts of a machine or mechanism and the forces which produce these motions



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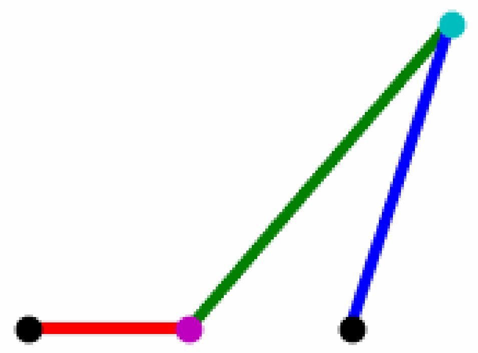

Contents of ENME608

Mechanism and Machine Concepts	Week 1
Linkages	Week 2
Position Analysis	Week 3 – 4
Velocity Analysis	Week 5
Acceleration Analysis	Week 6
Cam	Week 7
Cam (continued) and Gear	Week 8
Gear (continued)	Week 9
Dynamics Fundamentals	Week 10
Force Analysis	Week 11
Force Analysis (continued) and revision	Week 12

Mid-Sem Break

Contents of Mid-Semester Test

Mid-Sem Test

Contents of Final Exam

Final Exam (TBA)

Mechanisms and Machines

- A **mechanism** is a device which transforms motion to some desirable pattern and typically develops very low forces and transmits little power; A system of elements arranged to transmit motion in a predetermined fashion.

- A **machine** typically contains mechanisms which are designed to provide significant forces and transmit significant power
  - A machine has two functions:
    - – transmitting definite relative motion and;
    - – transmitting force.

  Motion and **Force** combined $\Rightarrow$ **Power**

Mechanisms and Machines (Cont.)

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Credit goes to https://pxhere.com/en/photo/606288

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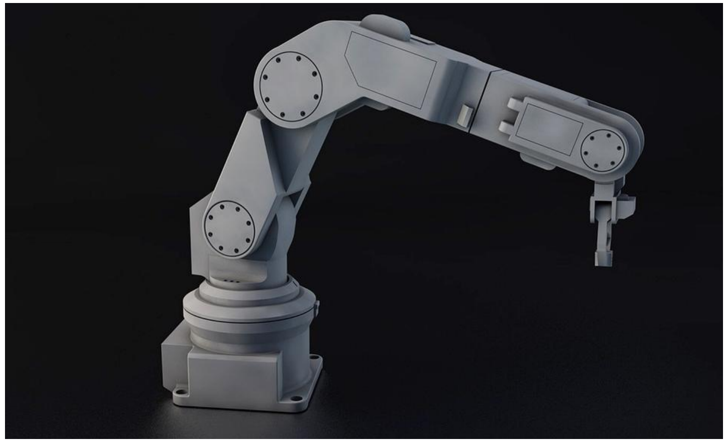

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Planar and Spatial Mechanisms

- In a **planar mechanisms**, all of the relative motions of the rigid bodies are in one plane or in parallel planes. (**2D**)
  - This course will mainly cover planar mechanisms.

- If there is any relative motion that is not in the same plane or in parallel planes, the mechanism is called the **spatial mechanism**. (**3D**)

Kinematics, Kinetics and Dynamics

- **Kinematics** of mechanisms is concerned with the motion of the parts without considering how the influencing factors (force and mass) affect the motion. Therefore, kinematics deals with the fundamental concepts of space and time and the quantities such as velocity and acceleration.

- **Kinetics** deals with action of forces on bodies. This is where the effects of gravity come into play.

- **Dynamics** is the combination of kinematics and kinetics.

Motions of Mechanisms

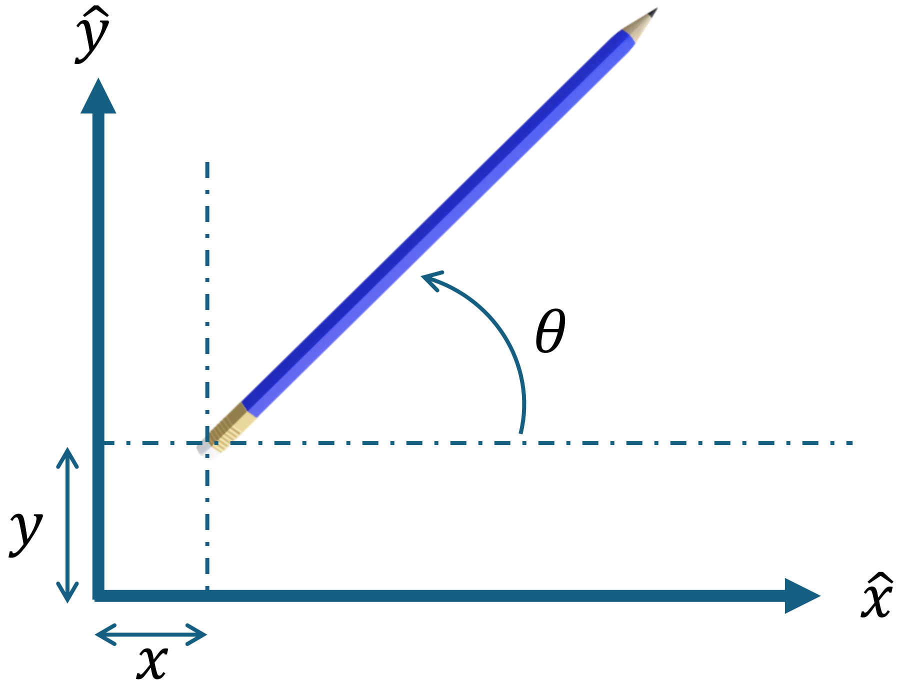
- **Degrees of Freedom (DOF)**

- **Types of Motions**
  - Pure rotation
  - Pure translation
  - Complex motion

- **Ways of Motion Transmission**
  - Intermediate links or connecting rods – linkages (links + joints)
  - Direct contact (e.g., cam and follower, gears)
  - Via a flexible connector (e.g., belt, chain)

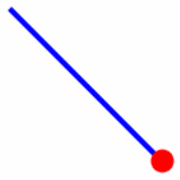
Motions of Mechanisms: DOF

A Mechanical system's DOF is equal to the number of independent parameters (measurements) which are needed to uniquely define its position in space at any instant of time.

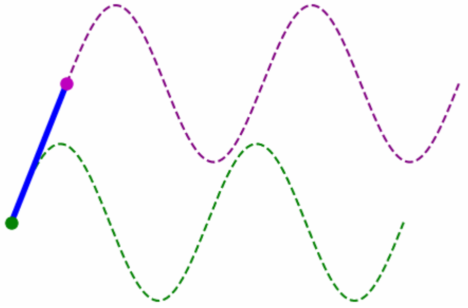


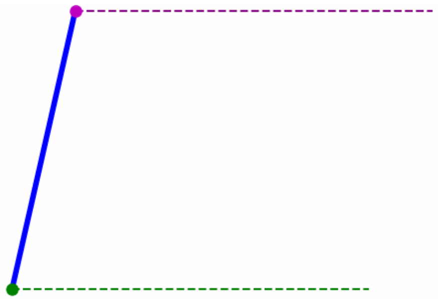
Motions of Mechanisms: Types of Motions

Pure rotation – the body possesses one point (centre of rotation) which has no motion. All other points on the body describe arcs about that centre.

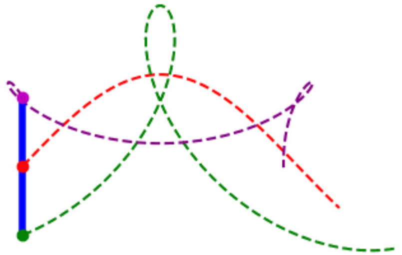


Pure translation – all points on the body describe parallel (curvilinear or rectilinear) paths.



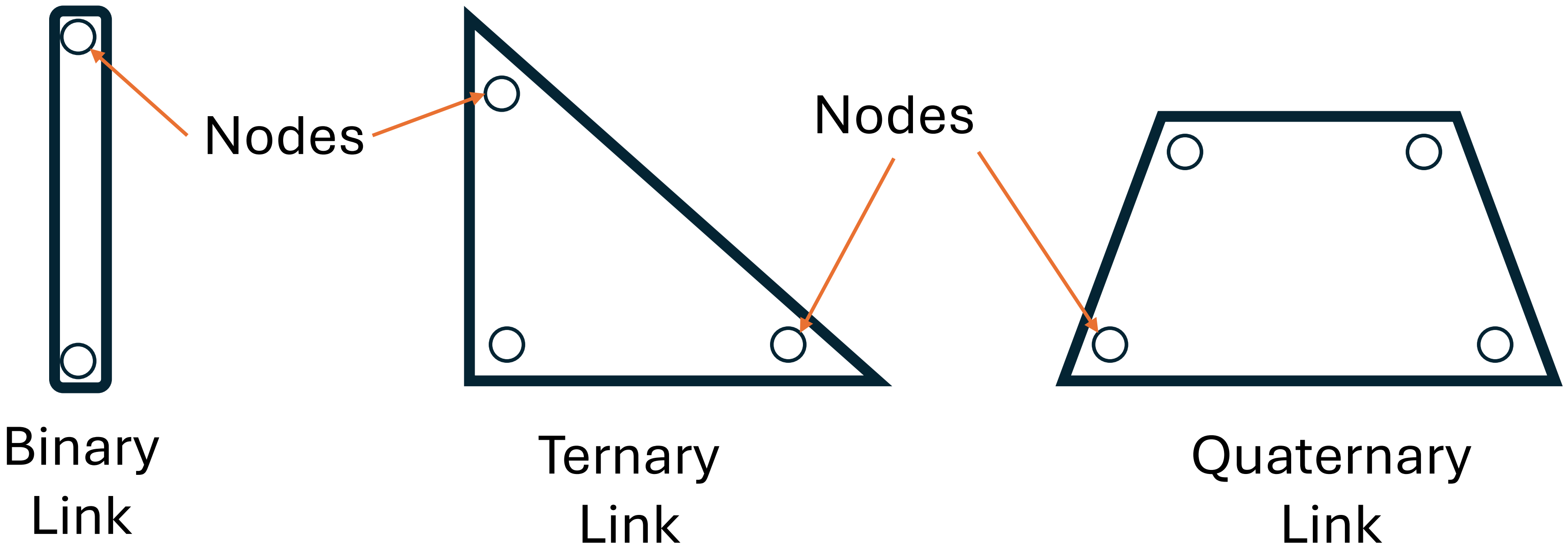


Complex motion – a simultaneous combination of rotation and translation



Motions of Mechanisms: Ways of Motion Transmission

- Intermediate links or connecting rods – **Linkages** (links + joints)
- **Link** - a rigid body having two or more pairing elements which connect it to other bodies for the purpose of transmitting force or motion



Motions of Mechanisms: Ways of Motion Transmission (Cont.)

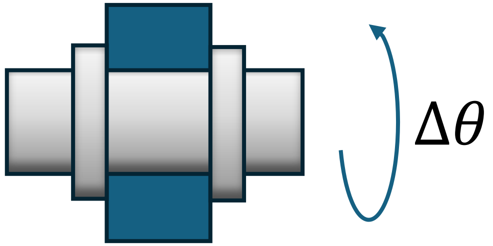
- **Joint (Pair)** – a connection between two or more links (at their nodes), which allows some motion, or potential motion, between the connected links.
- **Lower pair (full joint)** – a joint with surface contact (as with a pin surrounded by a hole).
- **Higher pair (half joint)** – a joint with point or line contact.
- **Joint** can also be classified by the number of DOF, physical closure or the number of links joined.

Motions of Mechanisms: Ways of Motion Transmission (Cont.)

More examples on Page 34, Chapter 2, Design of Machinery, 6 Edition

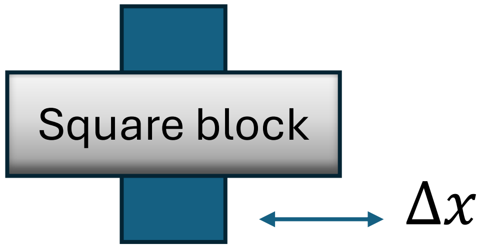
__ DOF

___ Pair



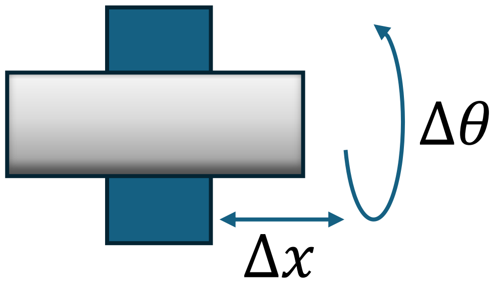
__ DOF

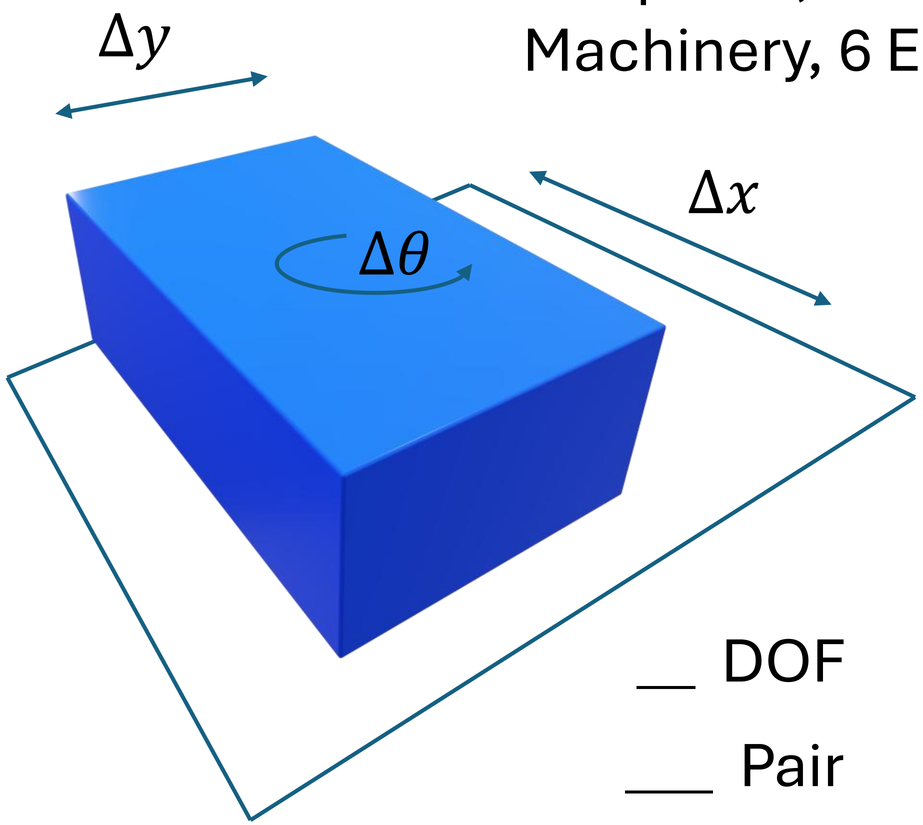
___ Pair



__ DOF

___ Pair





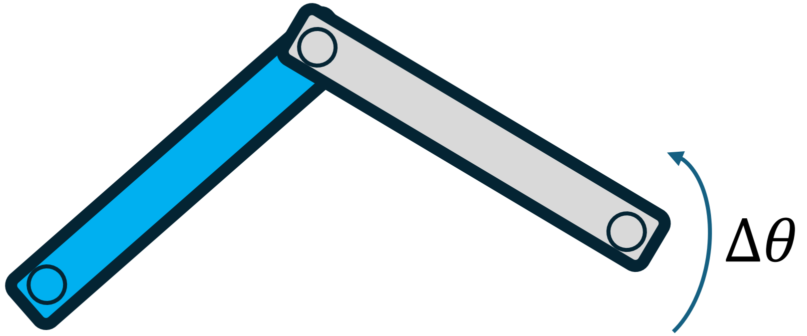
__ DOF

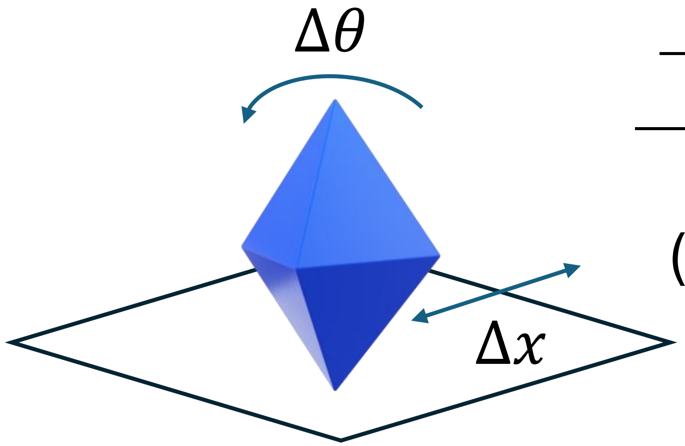
___ Pair

Motions of Mechanisms: Ways of Motion Transmission (Cont.)

__ DOF
___ Pair

Revolute (Pin) Joint



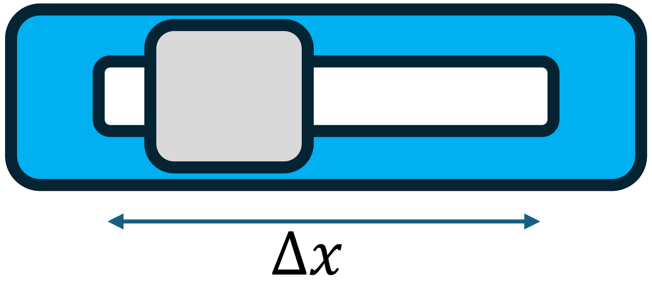


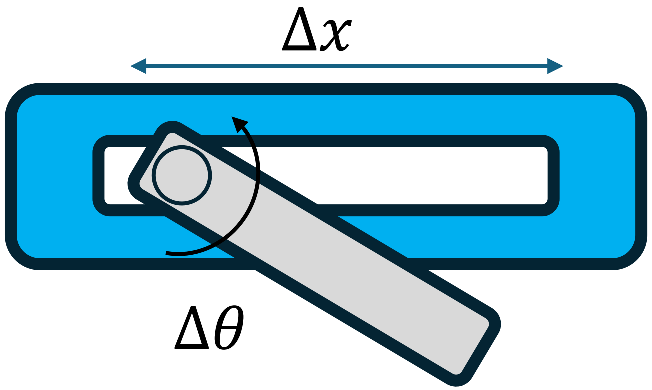
__ DOF
___ Pair

(Force closed)

__ DOF
___ Pair

Prismatic (Slider) Joint

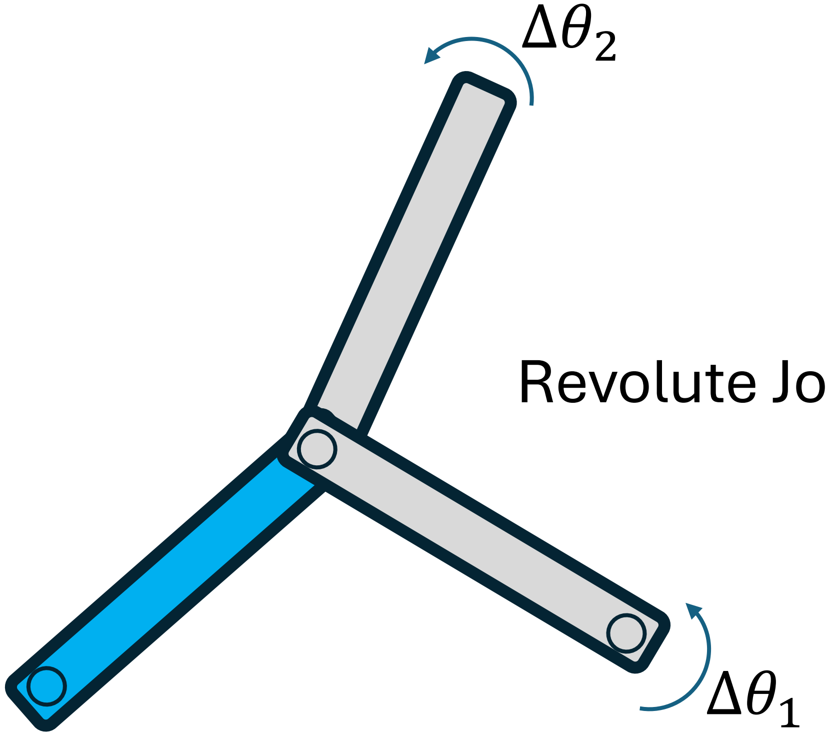




(Form closed)

__ DOF
___ Pair

Motions of Mechanisms: Ways of Motion Transmission (Cont.)



The order of a joint is __ less than the number of links joined.

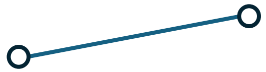
__ DOF

___ Pair

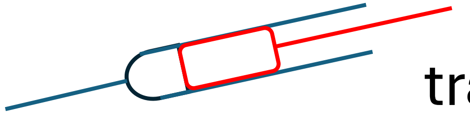
Kinematic Diagrams

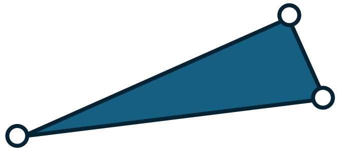
Kinematic link – a link between joints that allow relative motion between adjacent links.

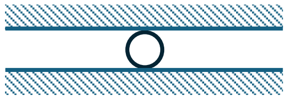
Schematic notions for common joints and links

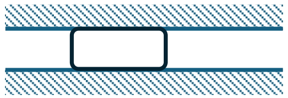

Binary Link

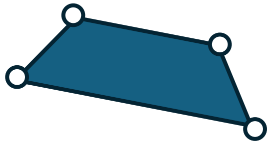

Moving rotating joint

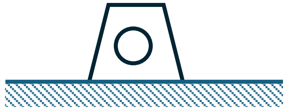

Moving translating joint

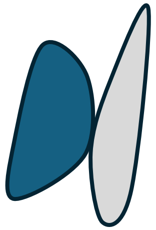

Ternary Link


Grounded half joint

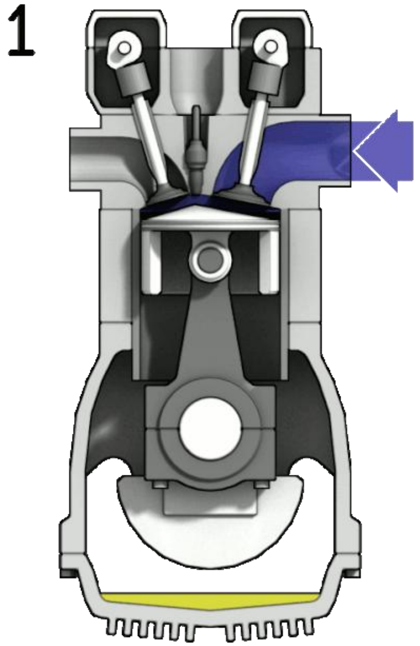

Grounded translating joint


Quaternary Link


Grounded rotating joint


Moving half joint

Kinematic Diagrams (Cont.)



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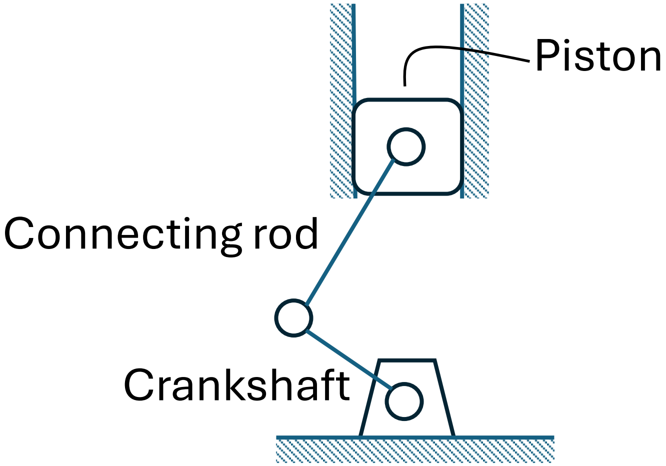
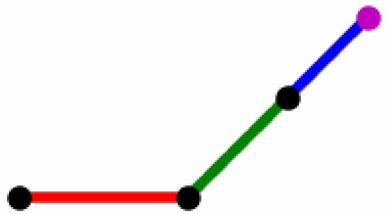


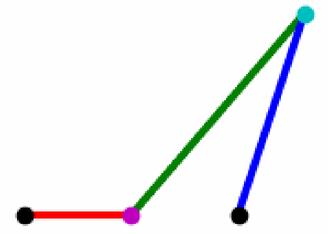
Kinematic chain, Mechanism and Machine

- **Kinematic chain** – An assemblage of links and joints, interconnected in a way to provide a controlled output motion.
- **Mechanism** – A kinematic chain in which at least one link has been "grounded," or attached, to the frame of reference.
- **Machine** – A combination of resistant bodies arranged to compel the mechanical forces of nature to do work accompanied by determinate motions.

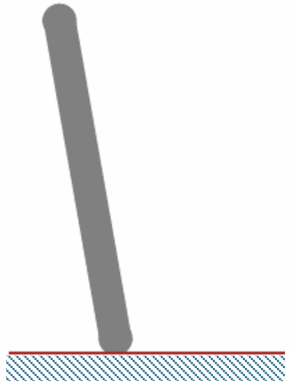
Kinematic chain, Mechanism and Machine (Cont.)

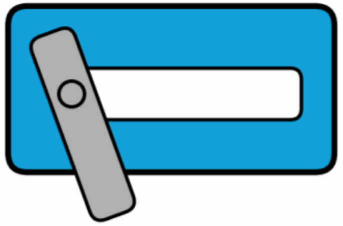
- **Open/closed kinematic chain (mechanism)** – a closed kinematic chain does not have open attachment points or nodes and may have one or more degrees of freedom.

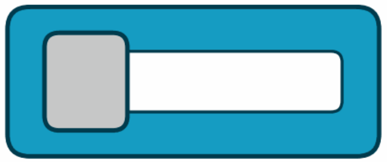




- **Dyad –** a special open mechanism consisting of two binary links and one joint.







Mobility or DOF of a planar mechanism

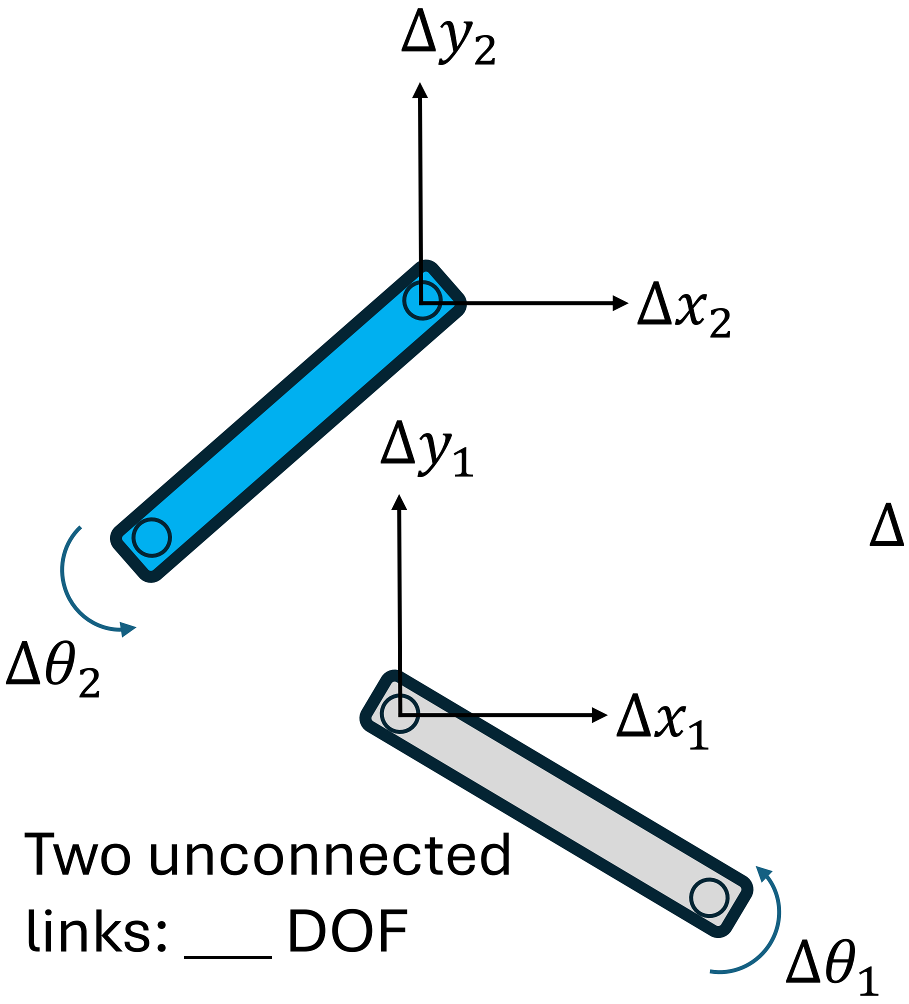
Grubler's (Kutzbach's) equation

$$M = 3(L-1) - 2J_1 - J_2 = 3(L-1) - 2(J_1 + 0.5J_2)$$

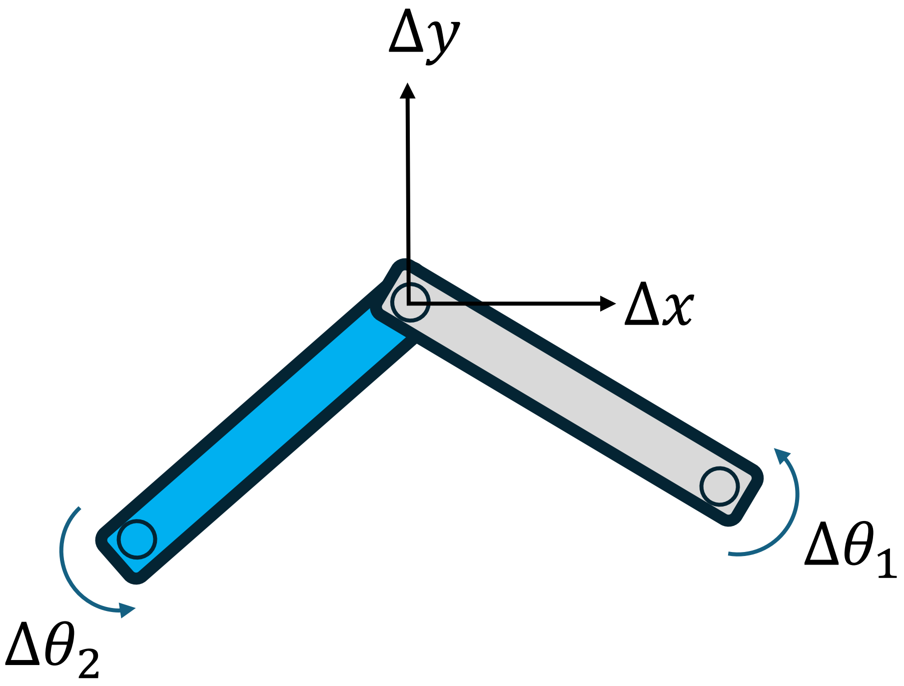
M : degree of freedom or mobility
L : number of links (including ground or frame)
J_1 : number of 1DOF joints (full joints)
J_2 : number of 2DOF joints (half joints)

$M \geq 1$: this device is a mechanism with M degrees of freedom
$M = 0$: this device is a statically determinate structure
$M \leq -1$: this device is a statically indeterminate structure or preload structure

Mobility or DOF of a planar mechanism (Cont.)

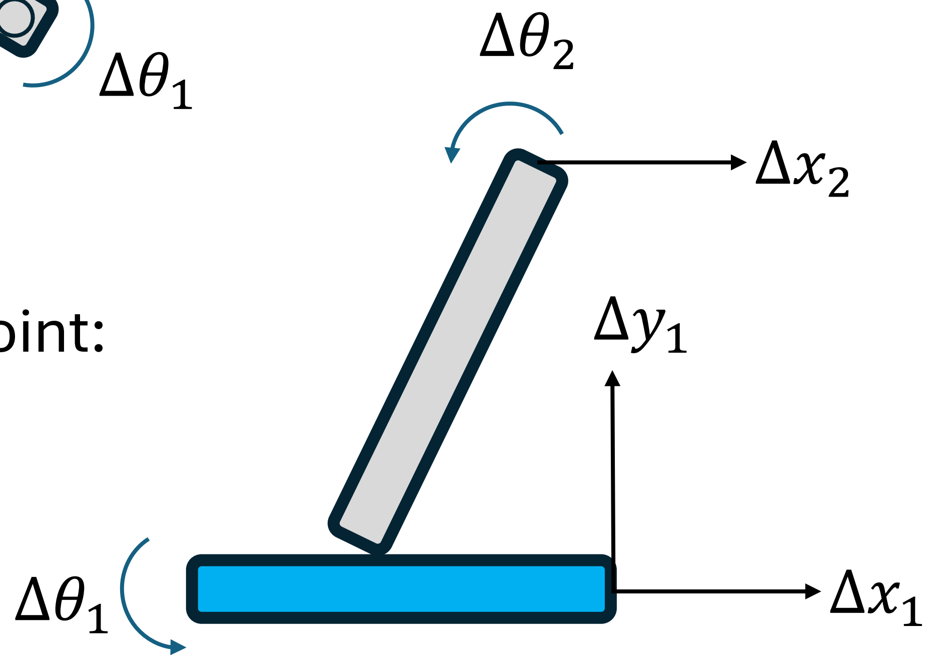


Two unconnected links: ___ DOF

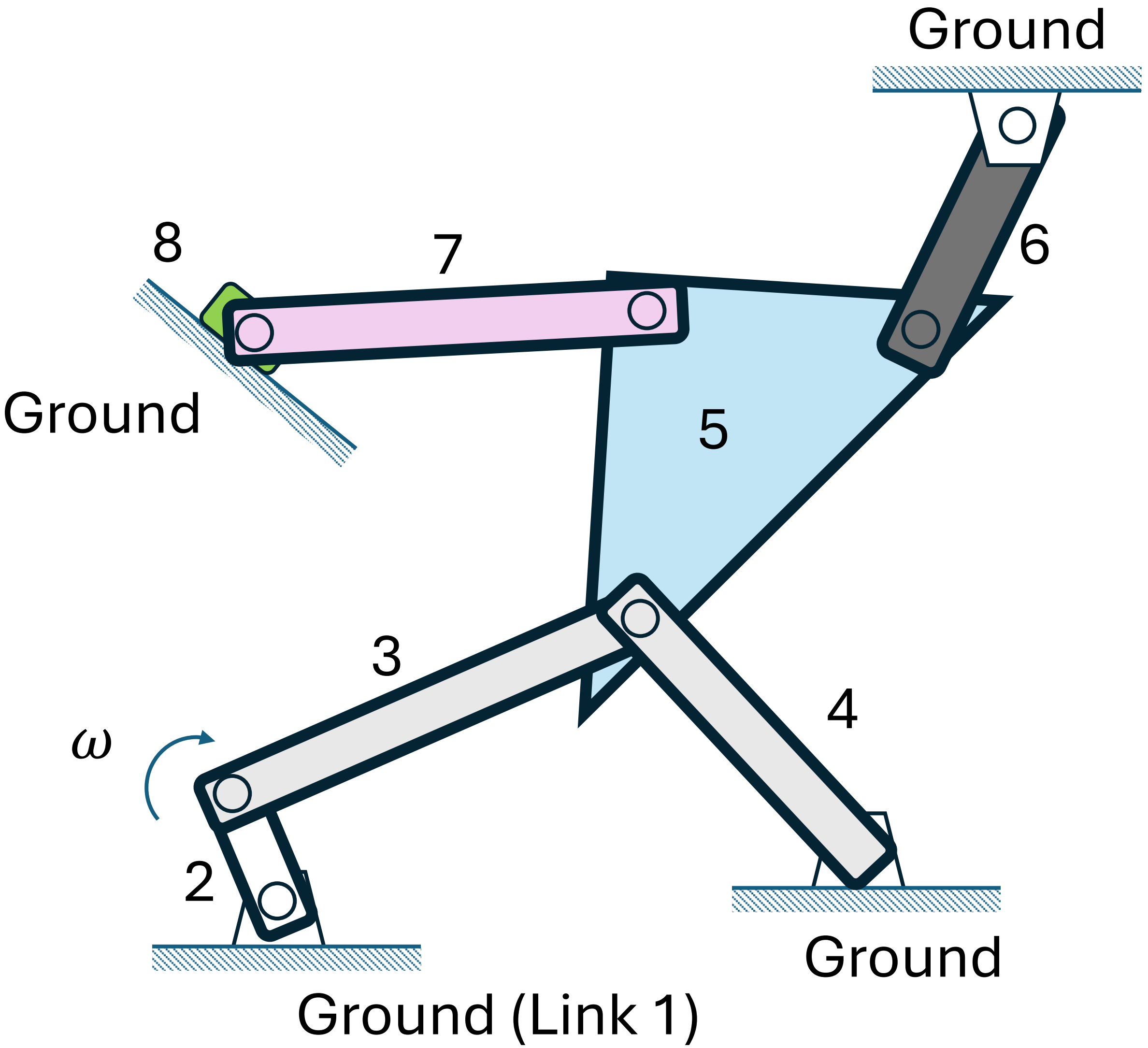


Two connected links connected by a full joint: ___ DOF

Connected links by a half joint: ___ DOF



Mobility or DOF of a planar mechanism (Cont.)



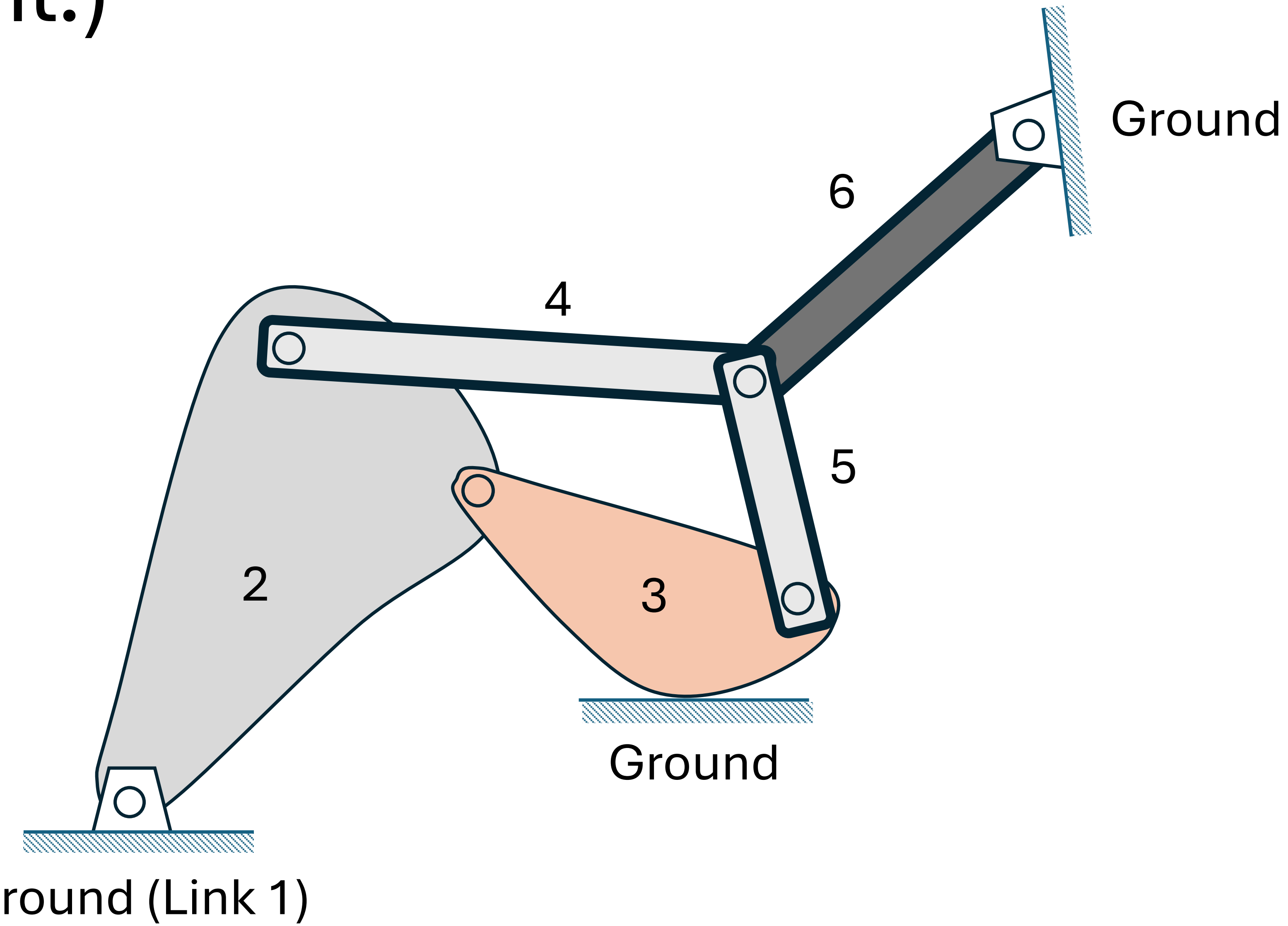
$L =$ ___

$J_1 =$ ___

$J_2 =$ ___

$M =$ ___

Mobility or DOF of a planar mechanism (Cont.)



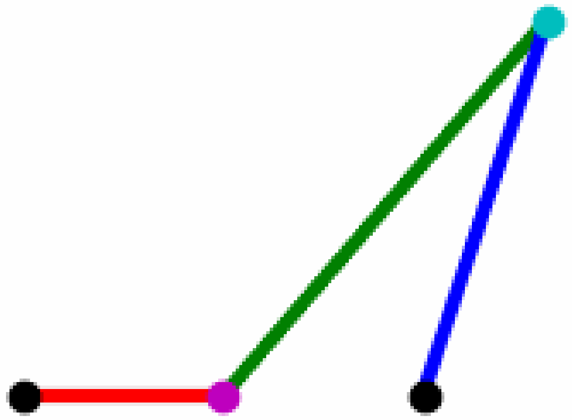
$L =$ ___

$J_1 =$ ___

$J_2 =$ ___

$M =$ ___

Mobility or DOF of a planar mechanism (Cont.)



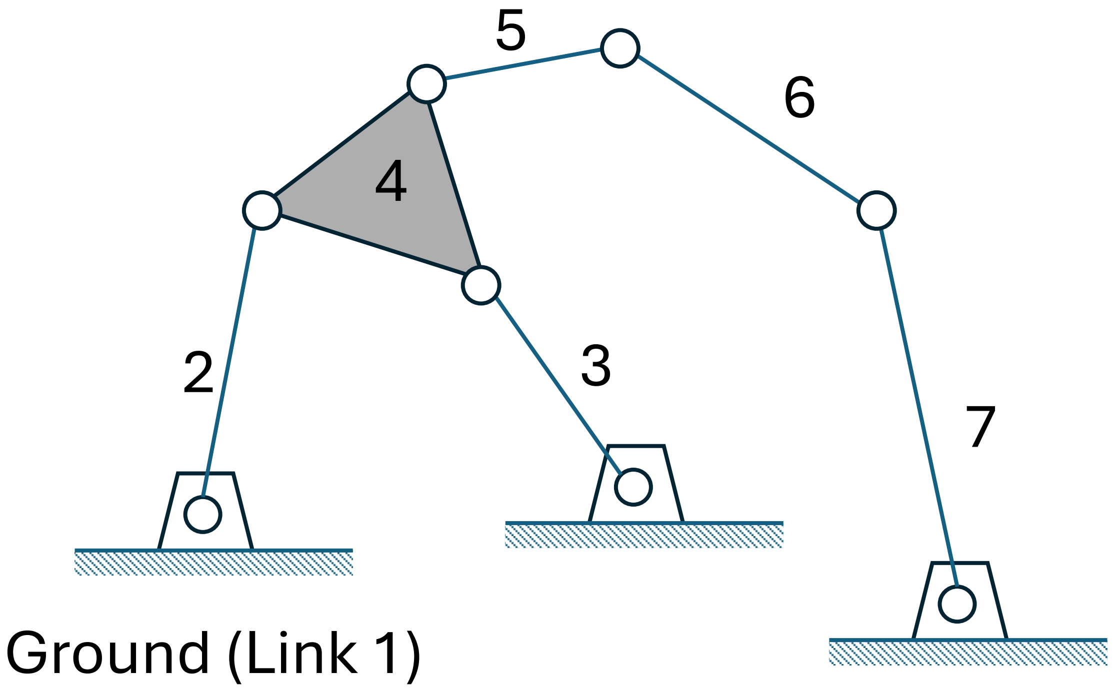
$L =$ ___

$J_1 =$ ___

$J_2 =$ ___

$M =$ ___

Mobility or DOF of a planar mechanism (Cont.)



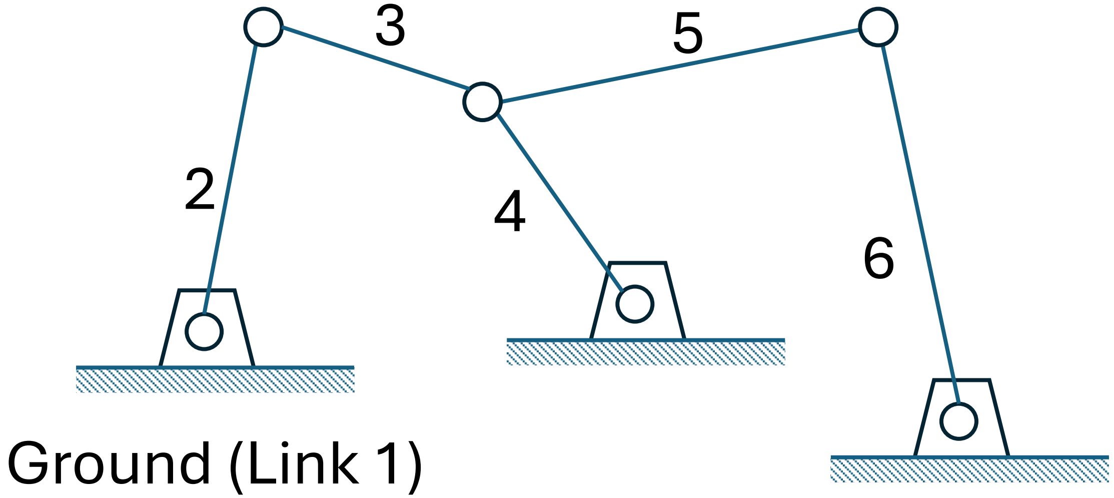
$L =$ ___

$J_1 =$ ___

$J_2 =$ ___

$M =$ ___

Mobility or DOF of a planar mechanism (Cont.)



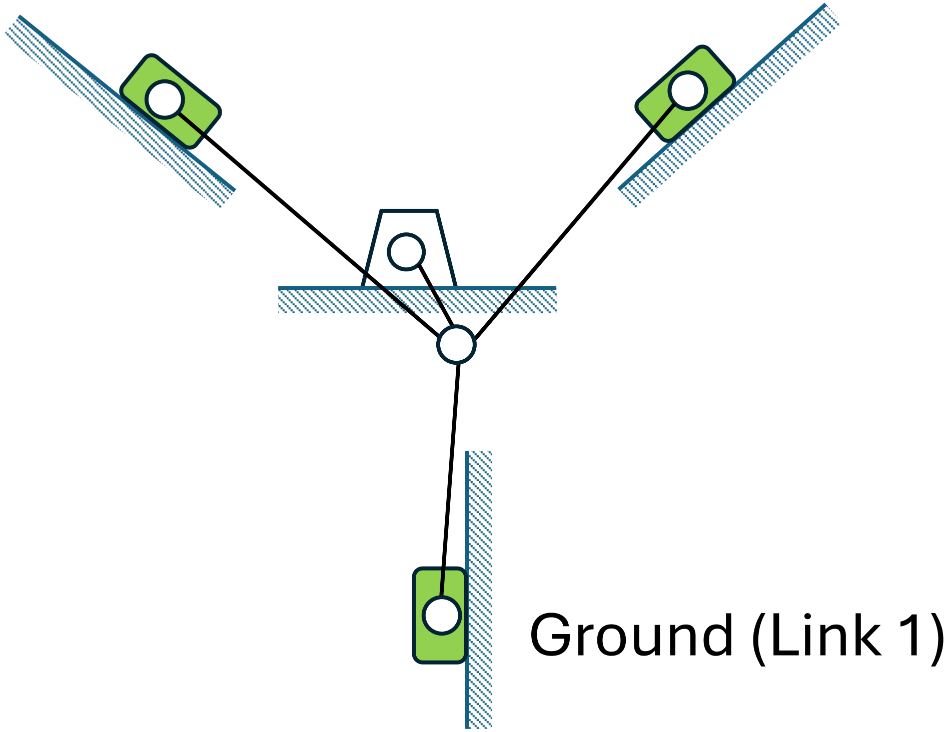
$L =$ ___

$J_1 =$ ___

$J_2 =$ ___

$M =$ ___

Mobility or DOF of a planar mechanism (Cont.)



$L =$ ___

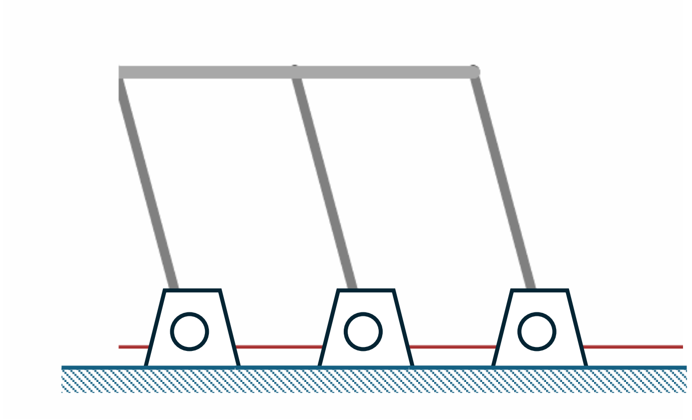
$J_1 =$ ___

$J_2 =$ ___

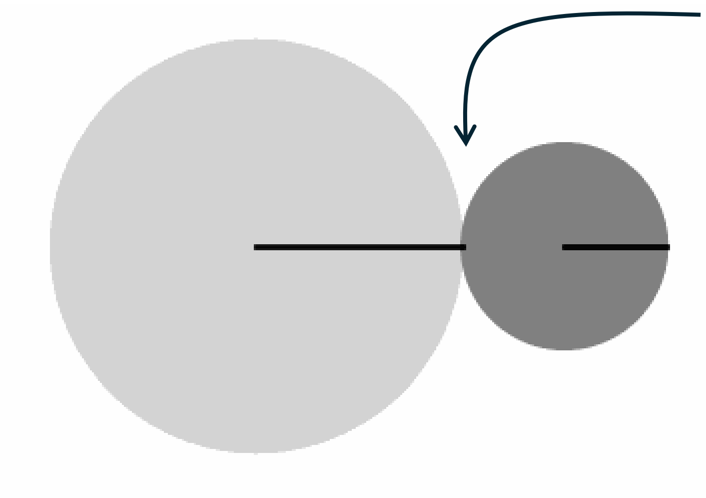
$M =$ ___

Paradoxes of Grubler criterion

E-quintet mechanism with DOF = 1 : disagrees with Grubler’s formula due to the unique geometry



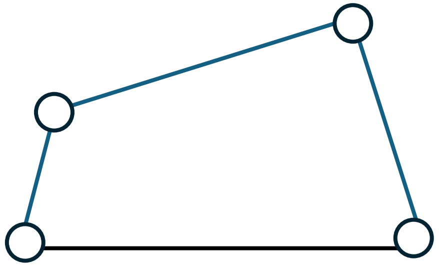
Rolling cylinders with DOF = 1 : disagrees with Grubler’s formula due to the unique geometry



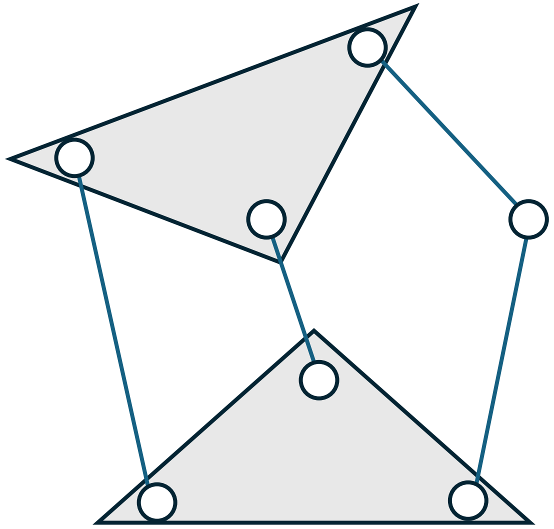
Full joint – pure rolling with no slip

Linkage Isomers

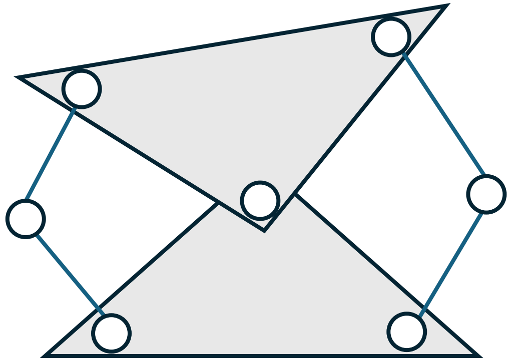
The links have various nodes to connect to other links' nodes to produce linkages of different motion properties.



The only fourbar isomer



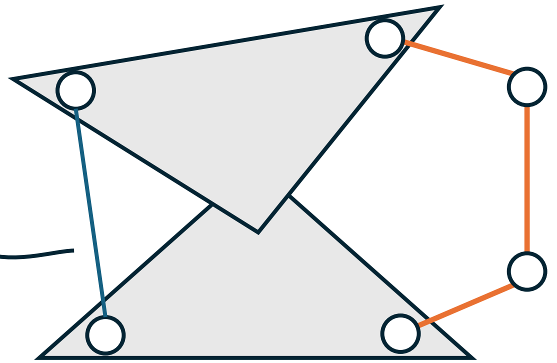
Stephenson's sixbar isomer



Watt's sixbar isomer

Linkage Isomers

Structural subchain reduces three links to a zero DOF "delta triplet" truss



Fourbar subchain concentrates the 1 DOF of the mechanism

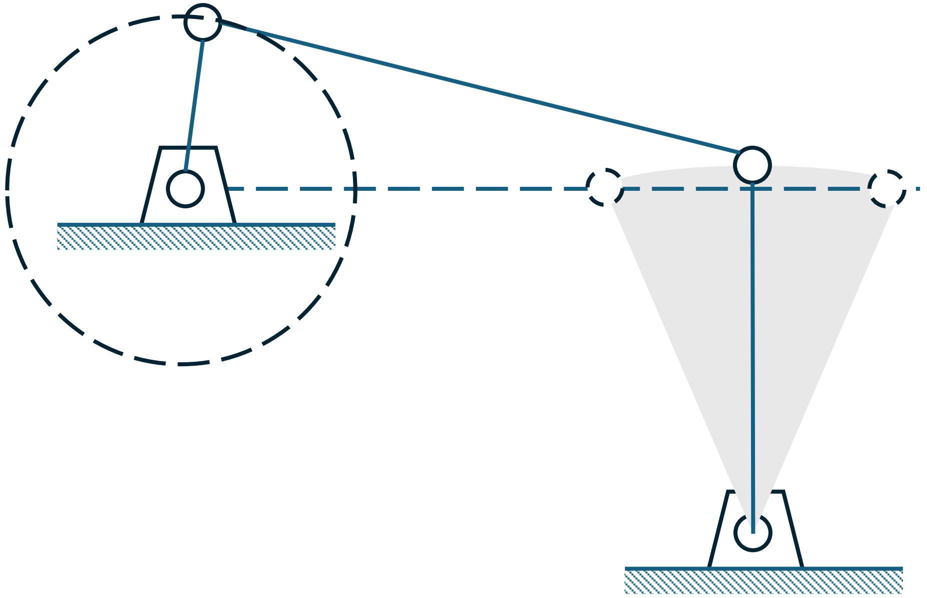
The invalid sixbar isomer which reduces to the simpler fourbar

Linkage Transformations

1. Revolute joints in any loop can be replaced by prismatic joints with no change in DOF of the mechanism, provided that at least two revolute joints remain in the loop.
2. Any full joint can be replaced by a half joint, but this will increase the DOF by one.
3. Removal of a link will reduce the DOF by one.
4. The combination of rules 2 and 3 above will keep the original DOF unchanged.
5. Any ternary or higher-order link can be partially “shrunk” to a lower-order link by coalescing nodes. This will create a multiple joint but will not change the DOF of the mechanism.
6. Complete shrinkage of a higher-order link is equivalent to its removal. joint will be created, and the DOF will be reduced. A multiple joint will be created, and the DOF will be reduced.

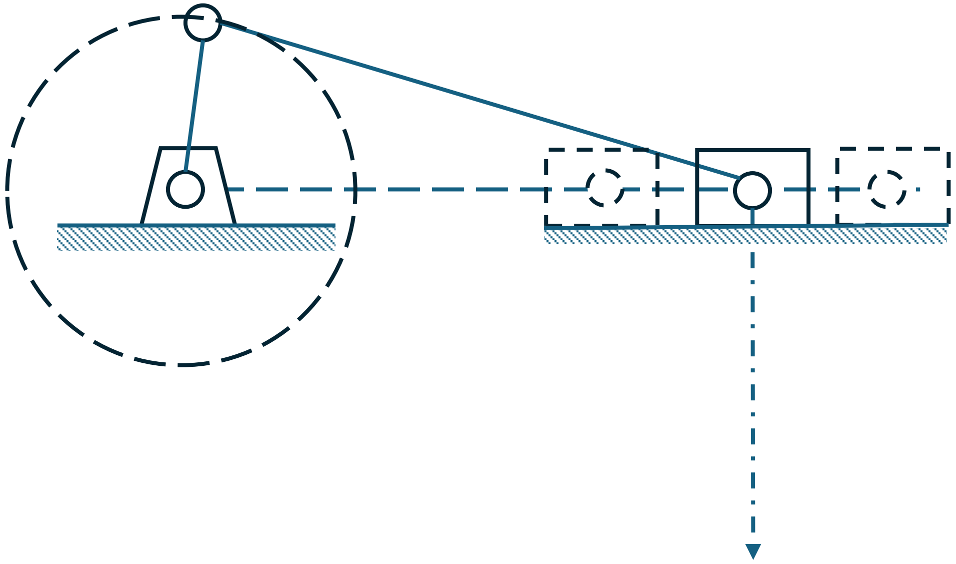
Linkage Transformations (Cont.)

Grashof crank-rocker



Rocker Pivot

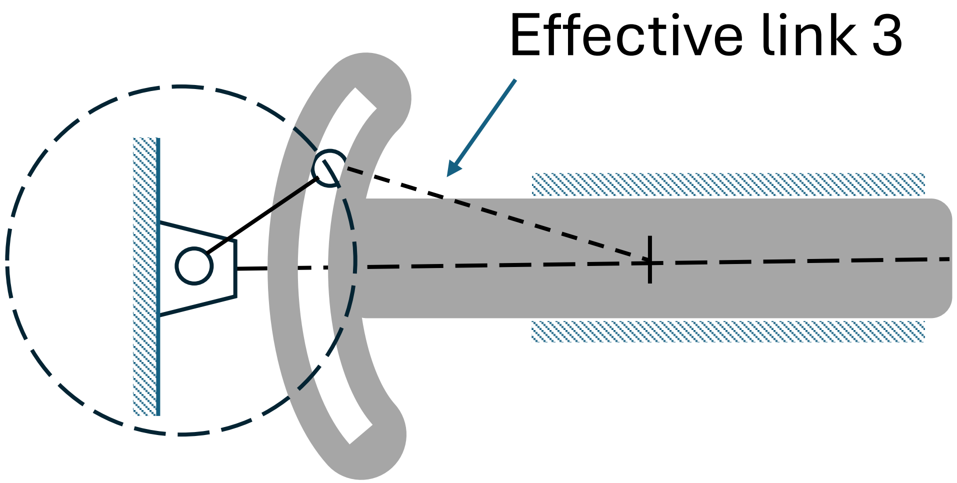
Grashof slider-crank



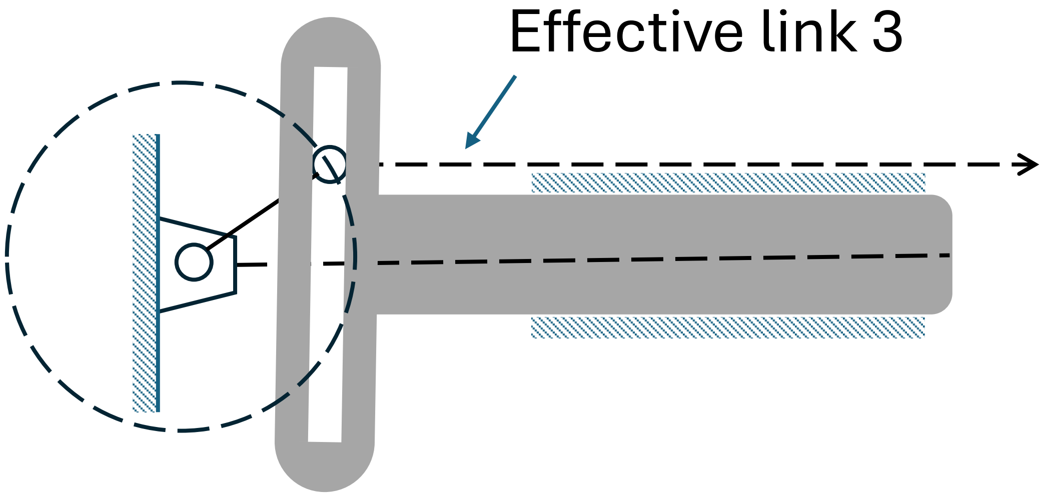
Effective rocker Pivot at Infinity

Linkage Transformations (Cont.)

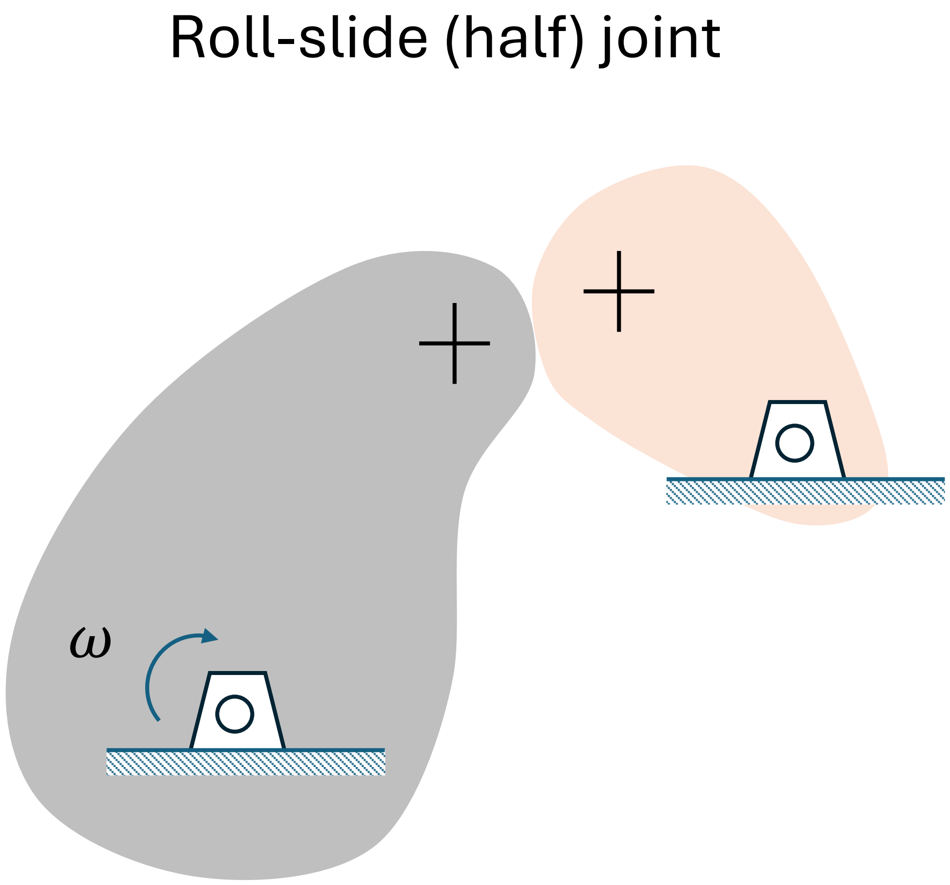
Slider-crank

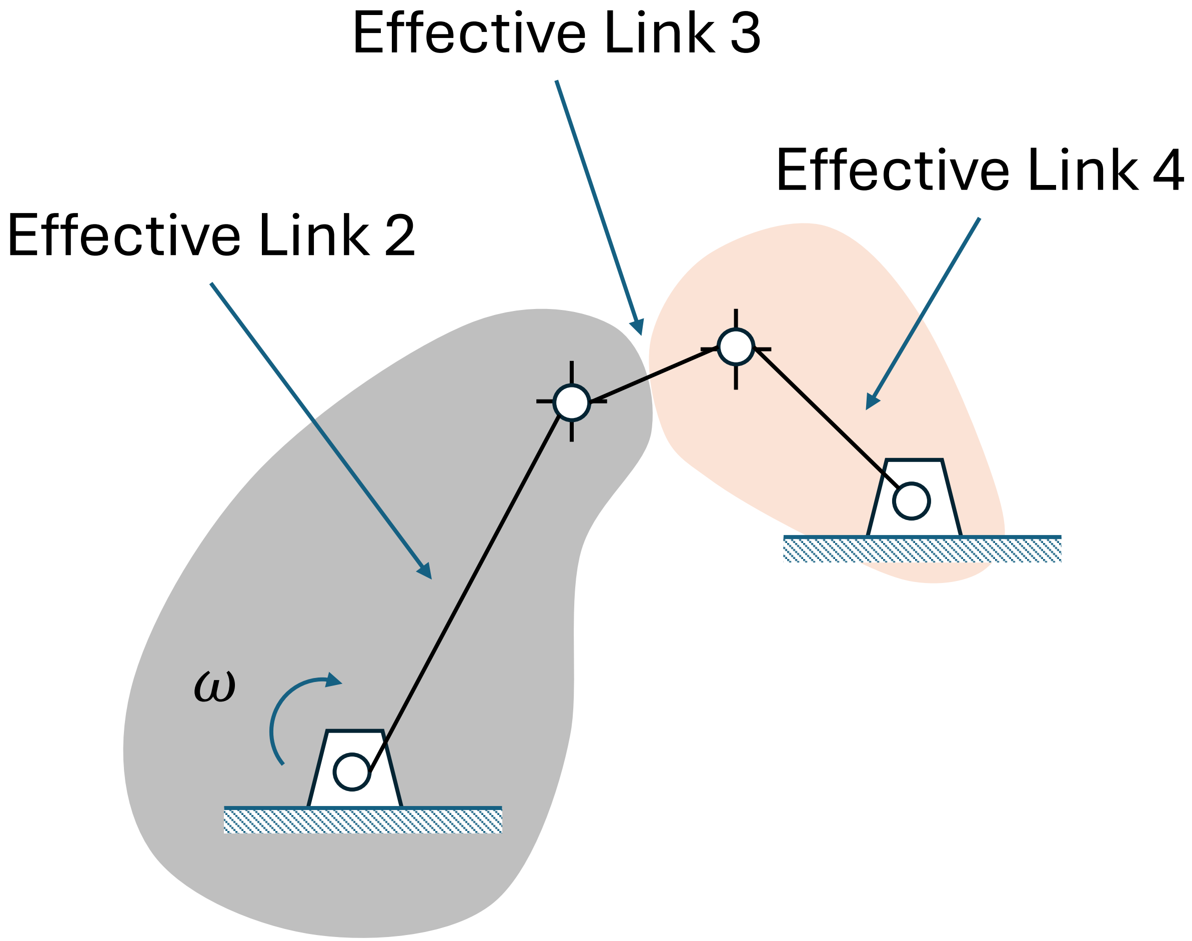


Scotch-yoke

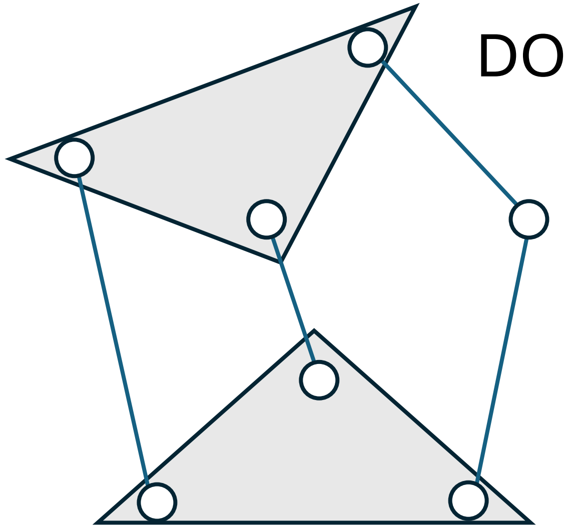


Linkage Transformations (Cont.)

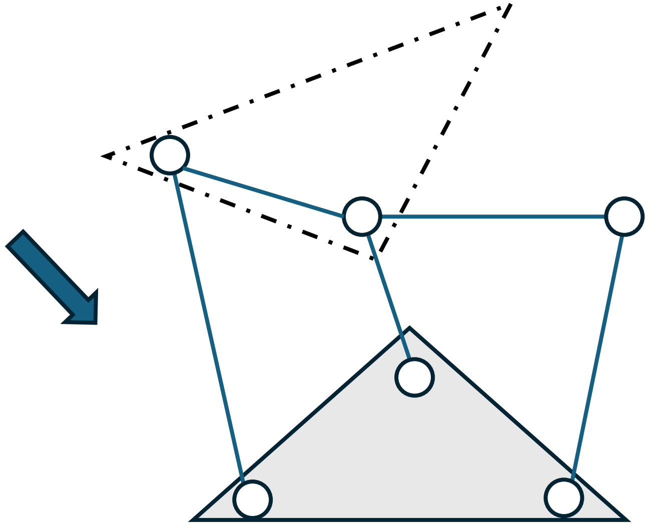




Linkage Transformations (Cont.)

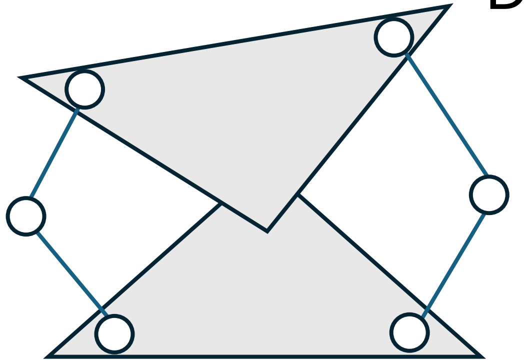


DOF = ____

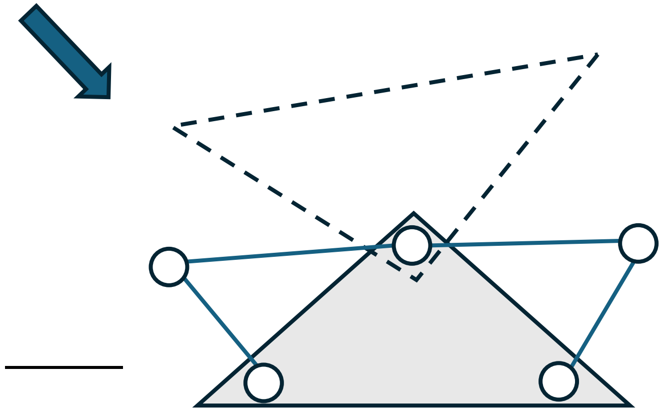


DOF = ____

Partial Shrinkage of higher link retains original DOF



DOF = ____

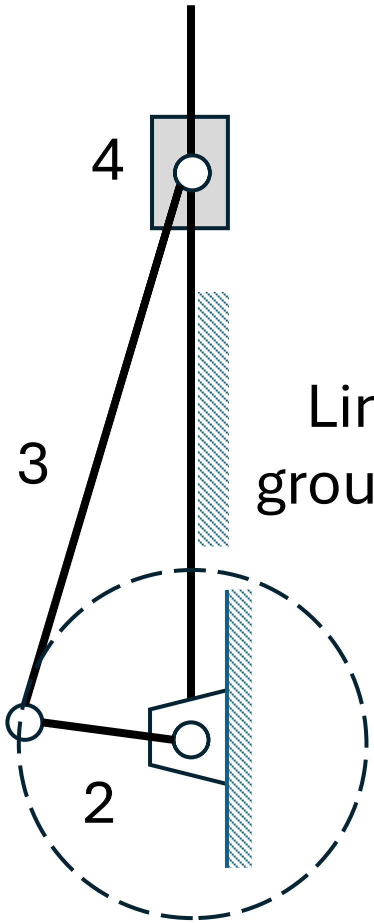


DOF = ____

Complete shrinkage of higher link reduces DOF by one

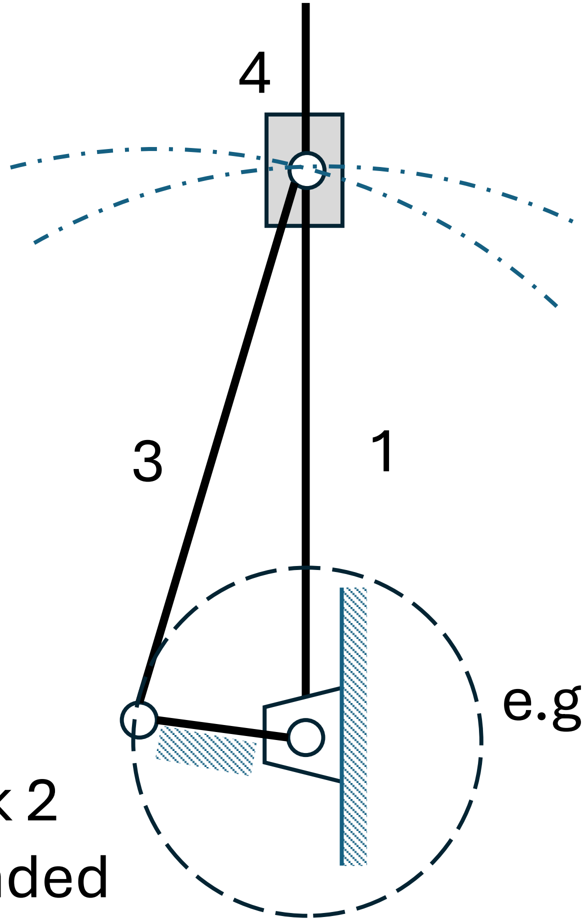
Inversion

By grounding a different link in the linkage



Link 1
grounded

Inversion #1
e.g., crank-slider,
engine piston

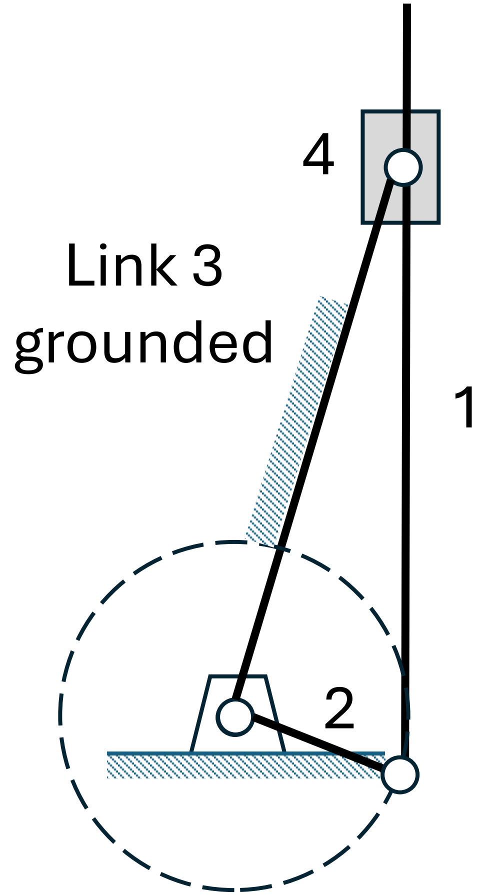


Link 2
grounded

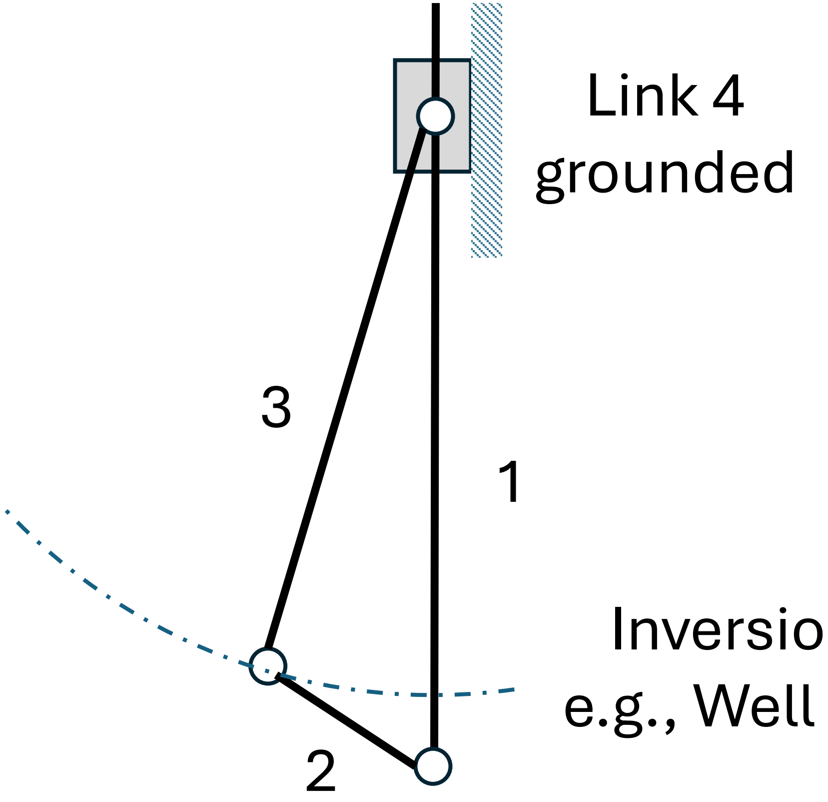
Inversion #2
e.g., crank-shaper,
Whitworth
mechanism

Inversion (Cont.)

By grounding a different link in the linkage



Inversion #3
e.g., Slider-crank



Inversion #4
e.g., Well pump